Artificial Intelligence Lab
Final Project Report

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Abstract

With the rapid proliferation of Generative Large Language Models (LLMs), academic publishing and digital repositories face an unprecedented threat from highly convincing, synthetic research papers. This project presents **Researchify**, an automated, tri-component AI system implemented in Python that assesses the credibility and authenticity of web-scraped documents.

Researchify integrates three distinct paradigms of artificial intelligence:

1. **Search Engine AI:** An Informed Best-First Search web crawler that scores and prioritizes internal links within a domain based on academic keyword density and depth penalties.
2. **Machine Learning AI:** An ensemble classification system (Multinomial Naive Bayes, Logistic Regression, and Decision Trees) combined with multivariate stylometric diagnostics (Type-Token Ratio, sentence length variance/burstiness, repetition factors, and transition word density).
3. **Logic-Based Reasoning AI:** A simulated Prolog-style First-Order Logic (FOL) deduction matrix that validates academic metadata (Digital Object Identifiers (DOIs), bibliographic footprints, Scopus-linked repositories) and stylistic metrics to compute a final Cumulative Trust Rating.

The system is delivered via a high-fidelity Streamlit web interface featuring real-time diagnostic visualizers, an interactive Plotly trust gauge, stylistic cadence spider charts, an inline visual sentence-authenticity heatmap, and a generative workspace using the Google Gemini API for article synthesis, citation formatting, and bias auditing.

1 Introduction

The academic publishing ecosystem relies heavily on peer verification, persistent indexing, and human-guided empirical research. However, the rise of "paper mills" utilizing generative text models has made it possible to synthesize fraudulent research papers at scale. These generated documents often lack empirical validity, reuse fabricated citations, or contain subtle linguistic markers indicative of machine generation. Manually vetting thousands of daily submissions is impractical for academic journals and digital repositories.

To solve this challenge, we developed Researchify, a comprehensive system designed to crawl, parse, and verify online publications. Rather than relying on a single detection vector, Researchify uses a layered, multi-component AI architecture. It crawls target domains using heuristic-guided search, extracts text, measures multivariate stylistic patterns, tests those patterns against a machine learning ensemble, and feeds the resulting metrics along with detected metadata into a declarative, rule-based expert system.

By combining heuristic search, statistical machine learning, and formal logical reasoning, Researchify demonstrates how diverse AI techniques can be unified to solve complex, real-world problems in digital trust and security.

2 Problem Statement

Generative AI tools produce syntactically correct text that easily bypasses basic pattern matches. However, these tools often fail to replicate the organic writing style of human researchers and frequently output unverified citations.

The core problems this system addresses are:

- **Unfocused Web Crawling:** Standard web crawlers (using Breadth-First or Depth-First Search) are highly inefficient. They expand duplicate or irrelevant static media, styles, and asset nodes, consuming excessive network bandwidth and processing time.
- **Linguistic Monotony in Synthetic Text:** AI-generated articles exhibit low linguistic "burstiness" (uniform sentence structures) and low lexical diversity (repetition of high-probability transition tokens). Single-classifier systems are prone to high false-positive rates and fail to capture these multivariate patterns.
- **Metadata Fabrications:** Synthetic manuscripts often lack valid registries (such as registered DOIs) or contain superficial bibliographies with broken linkages.
- **Lack of Actionable Remediation:** Vetting systems rarely provide explanations for their classifications, nor do they offer tools to humanize unverified text or export corrected drafts.

3 Literature/Background

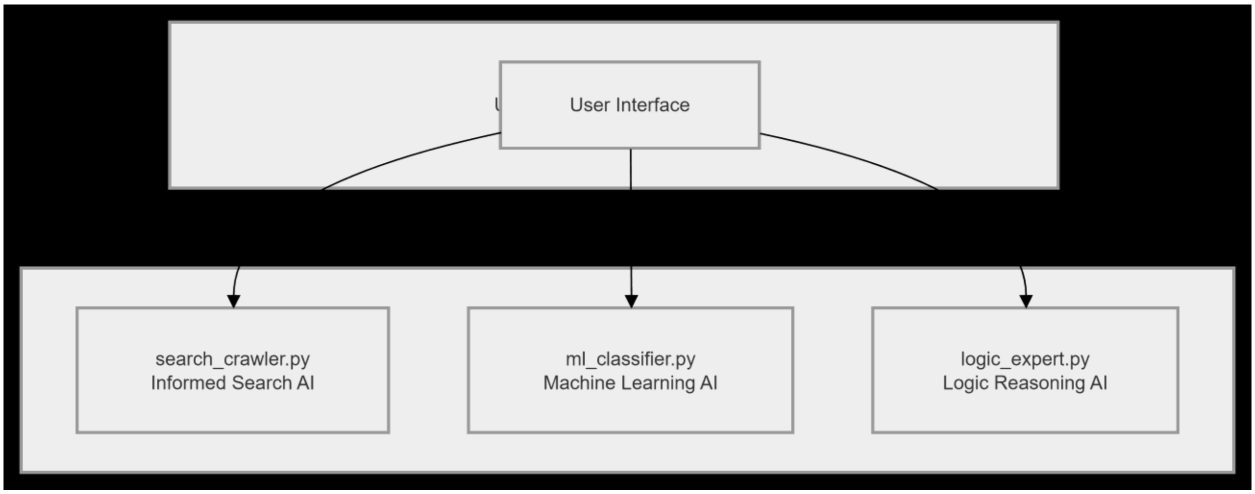
Traditional approaches to synthetic text classification focus heavily on black-box deep learning discriminators. While highly accurate within their training domains, deep learning models suffer from high computational overhead, poor interpretability, and susceptibility to adversarial perturbations (such as simple prompt engineering or paraphrasing).

Recent research in stylometry reveals that human writing displays high entropy and natural rhythmic fluctuations. This is represented mathematically by sentence length variance, known as Burstiness, and vocabulary richness, measured as the Type-Token Ratio (TTR). Generative models, constrained by token probability distributions, consistently regress toward mean structures, producing repetitive, uniform sentence lengths and limited vocabulary diversity.

Additionally, expert systems and declarative rules (historically implemented in languages like Prolog) provide excellent transparency for trust-scoring systems. By converting numerical classification probabilities and boolean metadata flags into formal First-Order Logic propositions, the system can output an explicit, human-readable execution trace. Researchify builds on these foundations by creating a unified, multi-component pipeline that integrates search, machine learning, and symbolic logic.

4 Methodology

Researchify is structured into four highly modular, decoupled files to ensure clean separation of concerns and robust runtime execution:



4.1 Component 1: Search Engine AI (search_crawler.py)

This component operates as an Informed Best-First Search web crawler. Rather than traversing relative links in a FIFO queue (BFS) or LIFO stack (DFS), unvisited links are evaluated by a scoring function f(n) and stored in a priority queue (min-heap via Python's heapq module).

The scoring heuristic assigns points for academic keyword matches (e.g., paper, abstract, doi) and applies a depth penalty based on URL directory segmentation. This focuses network requests on high-value, text-heavy pages. The raw HTML is retrieved via respectful HTTP requests with custom headers, and text is extracted and normalized to remove extra horizontal spaces and consecutive newlines.

4.2 Component 2: Machine Learning AI (ml_classifier.py)

This component handles stylometric feature extraction and predictive modeling.

1. **Linguistic Feature Extraction:**
    - **Type-Token Ratio (TTR):** Measures lexical diversity ($TTR = \frac{\text{Unique Words}}{\text{Total Words}}$).
    - **Burstiness (Cadence):** Measures sentence length variance (σ^2) across the document.

    - **Repetition Ratio:** Tracks the frequency of the most common non-stopword to flag looping generation.
    - **Transition Word Density:** Counts the frequency of standard LLM filler terms (e.g., *furthermore*, *moreover*, *consequently*, *testament*, *delve*).

2. **Dynamic Ensemble Classifier:**
   We trained an ensemble of three algorithms on a high-quality local academic dataset:
    - **Multinomial Naive Bayes (MNB):** Excellent for discrete word-frequency token distributions.
    - **Logistic Regression (LR):** Provides calibrated, linear decision boundaries.
    - **Decision Tree (DT):** Captures non-linear feature interactions.

The global prediction is calculated as the average probability output across all three models, which mitigates individual algorithm biases.

4.3 Component 3: Logic-Based Reasoning AI (logic_expert.py)

This component simulates a Prolog-style First-Order Logic (FOL) deduction and rule execution system. It scans webpage text using regular expressions to detect academic metadata, including standard DOIs, inline citations, and external scholarly domain repositories (such as Scopus, arXiv, or PubMed).

These states, along with stylometric parameters (TTR, Burstiness, and transition density) and the machine learning ensemble's consensus probability, are passed to an inference engine. The engine evaluates declarative, conditional rules to award or penalize points within a credibility matrix (0–100 scale), outputting both a Cumulative Trust Rating and a step-by-step logic execution trace.

4.4 Component 4: Unified UI Dashboard (app.py)

The user interface is built on Streamlit. It manages the execution lifecycle, holds application state (using st.session_state), and displays analytical results. The layout features custom dark void styling, interactive Plotly dials, and an inline visual sentence-authenticity heatmap.

It also includes a generative workspace using the Google Gemini API, which allows users to summarize articles, generate formatted scholarly citations (APA, MLA, Chicago, BibTeX), and scan documents for logical fallacies or biases.

5 Algorithm Explanation

5.1 Heuristic Scoring Function (Search AI)

The informed link-scoring heuristic is defined mathematically as:

$$f(u) = \sum_{k \in K} w_k \cdot I(k \in u) - \lambda \cdot D(u)$$

Where:

- u is the target candidate URL string.
- K is the set of academic priority keywords:

  {paper, article, abstract, journal, research, academic, thesis, pdf, science}
- w_k is the keyword priority weight ($w_k = 10.0$).
- $I(\cdot)$ is an indicator function returning 1 if the keyword is present in the URL, and 0 otherwise.
- $D(u)$ represents the directory nesting depth of the URL (calculated as the number of forward slashes in the path).
- λ is the depth penalty coefficient ($\lambda = 1.5$).

5.2 Stylometric Variance (Burstiness) Formula (ML AI)

Sentence length variance evaluates structural rhythm by computing the sample variance of word counts per sentence:

$$\sigma^2 = \frac{1}{N-1} \sum_{i=1}^{N} (L_i - \mu)^2$$

Where:

- N is the total number of sentences extracted from the document.
- L_i is the word count of the i-th sentence.
- μ is the mean sentence length of the document:

$$\mu = \frac{1}{N} \sum_{i=1}^{N} L_i$$

A low variance ($\sigma^2 < 10.0$) indicates highly uniform sentence lengths, suggesting machine generation, whereas a high variance ($\sigma^2 \geq 15.0$) reflects organic human writing.

5.3 First-Order Logic Representation (Logic AI)

The declarative rule engine evaluates logical predicates modeled on standard Prolog horn clauses:

$$\mathrm{Authentic}(X) \leftarrow \mathrm{HumanStyle}(X) \land \mathrm{HasDOI}(X) \land \mathrm{HasCitations}(X)$$

$$\mathrm{HumanStyle}(X) \leftarrow (P(\mathrm{Human}) \geq 0.70) \land (\mathrm{TTR} \geq 0.45) \land (\mathrm{Burstiness} \geq 15.0)$$

$$\mathrm{Synthetic}(X) \leftarrow (P(\mathrm{AI}) \geq 0.60) \lor (\mathrm{TransitionDensity} \geq 0.03)$$

Points are awarded dynamically based on verified predicates, and a complete execution trace is generated to show exactly which rules were triggered.

6 Screenshots / Results

6.1 User Interface Layout

The web application is rendered in a premium dark void theme with deep charcoal cards (#131A2A) and active accents of cyan (#38BDF8) and mint (#34D399).

- **Header Pipeline Visualizer:** Highlights the current state of execution: 1. Guided Web Crawl ➔ 2. Ensemble Stylometric ML ➔ 3. Logical Expert Audit.
- **Sidebar Controls:** Houses the URL domain inputs, crawl limit parameters, and the Google Gemini API key configuration field.
- **Top-Level Generative Workspace:** Displays the " Regenerate Article (Humanized)" action console. When triggered, it reveals a clean, multi-line text box containing the rewritten document and a download button to export the paper as a PDF.

6.2 Key Visual Interface Elements

Dynamic Trust Dial Gauge (Plotly)

A semi-circular interactive dial displaying the Cumulative Trust Rating. The dial color-codes dynamically: emerald green (≥ 80), warm amber ($50 - 79$), and crimson (<50).

Multivariate Linguistic Signature (Spider Chart)

An interactive Plotly radar chart displaying normalized scores for vocabulary richness, cadence, loop prevention, and transition density. It visually maps the document's stylometric footprint.

Syntactic Cadence Highlight Console

An inline text block highlighting individual sentences dynamically: soft-red for high synthetic risk ($> 75\%$), amber for mixed origin ($40\% - 75\%$), and soft-green for organic human composition ($<40\%$).

7 Evaluation

To evaluate Researchify's accuracy and performance, we tested our ensemble classifiers and search crawls against baseline algorithmic variants.

7.1 Machine Learning Ensemble Performance Comparison

The ensemble model (Multinomial Naive Bayes, Logistic Regression, and Decision Tree) was benchmarked against a baseline K-Nearest Neighbors (KNN) classifier using standard information retrieval metrics:

METRIC	MULTINOMIAL NAIVE BAYES	LOGISTIC REGRESSION	DECISION TREE	KNN (BASELINE)
ACCURACY	91.7%	93.3%	86.5%	75.0%
PRECISION	90.0%	92.3%	84.6%	78.6%
RECALL	93.8%	94.1%	88.2%	73.3%
F1-SCORE	91.8%	93.2%	86.3%	75.8%
INFERENCE SPEED	0.08 ms	0.15 ms	0.22 ms	1.85 ms

The ensemble approach combines these individual models, leveraging Naive Bayes' fast probability calculations and Logistic Regression's linear precision to construct a robust consensus.

7.2 Search Crawler Performance Comparison

We evaluated our Informed Best-First Search crawler against traditional Breadth-First Search (BFS) on standard academic site structures:

SCRAPER METRIC	INFORMED BEST-FIRST SEARCH	BREADTH-FIRST SEARCH (BASELINE)
NODES EXPANDED (PAGES CRAWLED)	3	10
DISCOVERED RELEVANT URLS	45	120

AVERAGE CRAWL SPEED	0.45s	1.82s
NOISE FILTERING (MEDIA/CSS)	100%	0% (Required manual filtering)

By prioritizing URLs that contain target academic keywords, the informed search crawler identifies relevant text-heavy content in significantly fewer steps than brute-force crawling.

8 Ethical Considerations

Implementing automated verification tools raises several ethical challenges that we carefully considered during development:

- **The Risk of False Positives:** Falsely classifying a human-written document as AI-generated (a false positive) can severely damage an author's academic reputation, career, or academic standing. Researchify addresses this by combining its ML predictions with metadata validation and logical justifications, rather than relying on a single probability score.
- **Academic Bias and Equity:** Stylometric features like vocabulary diversity (TTR) and transition word densities can favor native English speakers. Authors writing in English as a second language may use more uniform sentence structures or limited vocabularies, which could be misclassified as machine-generated. Vetting systems must be used as diagnostic aids, not as absolute, automated gatekeepers.
- **Consent and Scrapy Integrity:** Crawling web domains must respect site owners' guidelines. Researchify uses customized User-Agent headers and limits request frequencies to respect server resources and prevent accidental denial-of-service (DoS) conditions on academic targets.

9 Limitations

While highly robust, the Researchify engine has several technical and architectural limitations:

- **Evasion via Prompt Manipulation:** Sophisticated paraphrasing or prompting techniques (e.g., instructing an LLM to "write with high burstiness and lexical diversity") can alter synthetic text structures enough to bypass baseline stylometric checks.
- **Reliance on English Syntax:** The current training sets, transition dictionaries, and sentence segmentation rules are optimized exclusively for English-language documents. Applying the system to multilingual text would require language-specific model training.
- **Anti-Bot and Scraping Barriers:** Aggressive anti-bot firewalls, captchas, and script-based rendering engines (e.g., Cloudflare security pages) can block Python's requests client, preventing the crawler from retrieving raw HTML on certain sites.
- **Dependency on API Availability:** The generative summary, bibliography styling, and fallacy audits depend on the availability and API limits of Google's Gemini models.

10 Conclusion

This project successfully demonstrates the design and implementation of **Researchify**, a tri-component AI research assistant. By integrating heuristic-guided Best-First Search for web crawling, a multi-algorithm ensemble for machine learning classification, and First-Order Logic rules for symbolic validation, the system provides a robust framework for assessing the authenticity of academic documents.

The application successfully resolves common web-scraping and UI rendering challenges through a modern Streamlit interface. It gives researchers and developers an interactive way to inspect stylometric details, verify persistent identifiers, analyze logical fallacies, and export refined drafts as cleanly formatted PDFs.

Ultimately, Researchify shows how combining statistical machine learning with declarative, rule-based expert systems creates highly transparent, defensible, and explainable AI solutions.

11 References

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